

COMP C0835

Object Oriented Systems

Week 8 Lecture

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Outline

- Understanding Systems Design
- Designing the User Interface
- The User Experience and the User Interface design
- The Fundamental Principles of User-Interface Design
- Transitioning from Analysis to User-Interface Design
- Dialogs and Storyboards in user interface design
- Understanding the Interface Design Fundamentals

Learning Objectives

- Understand the difference between Systems Analysis & Systems Design in IT projects
- Explain the concepts of user experience, user interface, and usability
- Describe the metaphors that can be used to assist in user-interface design
- Describe important characteristics of human interface objects that affect usability
- Understand the Interface Design Fundamentals
- Discuss the important principles of navigation through a software application

Learning Objectives (contd.)

- Understand the Interface Design Fundamentals
- Understand use case specification in system development
- Describe important guidelines in user-interface design for desktop applications
- Describe important guidelines in user-interface design for mobile devices
- Design printed and on-screen reports appropriate to users' needs

User Interface Design

- User interface design is the design of user interfaces for systems, or mobile devices, with the intent to maximize usability and user experience
- Good user interfaces are based on good design principles – visibility, affordance, feedback, etc.
- Poorly designed user interface can make the information system unusable

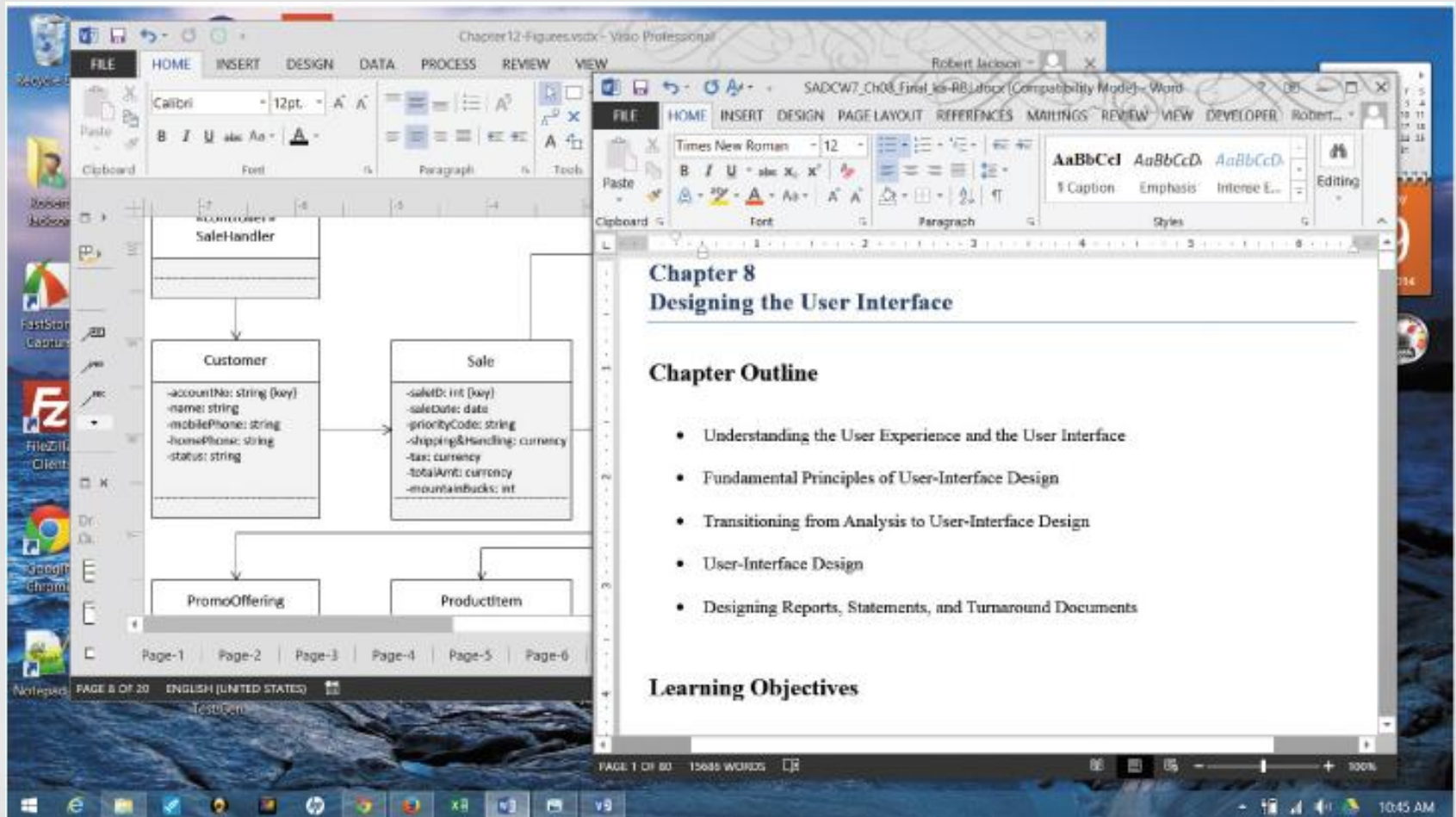
User Experience

- User Interface – inputs and outputs that directly involve a human user/actor
 - A dialog goes on between actor and system
- User interface design must focus on entire User Experience (Human Computer Interaction –HCI)
- Called User-Centered Design
 - Focus early on users and their work
 - Evaluate designs to ensure usability
 - Use iterative development
- Usability is the objective (User friendliness)

Metaphors of Human Computer Interaction

Metaphor	Description	Example
Direct manipulation	Manipulating objects on a display that look like physical objects (pictures) or that represent them (icons)	The user drags a folder icon to an image of a recycle bin or trash can to delete a collection of files.
Desktop	Organizing visual display into distinct regions, with a large empty workspace in the middle and a collection of tool icons around the perimeter	At computer startup, a Windows user sees a desktop, with icons for a clock, calendar, notepad, inbox and sticky notes (the computer interface version of a physical Post-It note).
Document	Visually representing the data in files as paper pages or forms; these pages can be linked together by references (hyperlinks)	The user fills in a form field for a product he or she owns, and the manufacturer's Web site finds and displays the product's manual as an Adobe Acrobat file, which contains a hyperlinked table of contents and embedded links to related documents.
Dialog	The user and computer accomplishing a task by engaging in a conversation or dialog by using text, voice, or tools, such as labeled buttons	The user clicks a button labeled "troubleshoot" because the printer isn't working. The computer prints questions on the display, and the user responds by typing answers or selecting responses from a printed list.

Direct Manipulation, Desktop, and Document Metaphors On One Screen



User experience from the user interface (HCI)



Principles of User-Interface Design

Principles of User-Interface Design

- Human-Interface Objects
 - Affordance
 - Visibility
- Consistency
- Discoverability
- Dialog Closure
- Readability & Navigation
- Usability & Efficiency

Principles of User-Interface Design

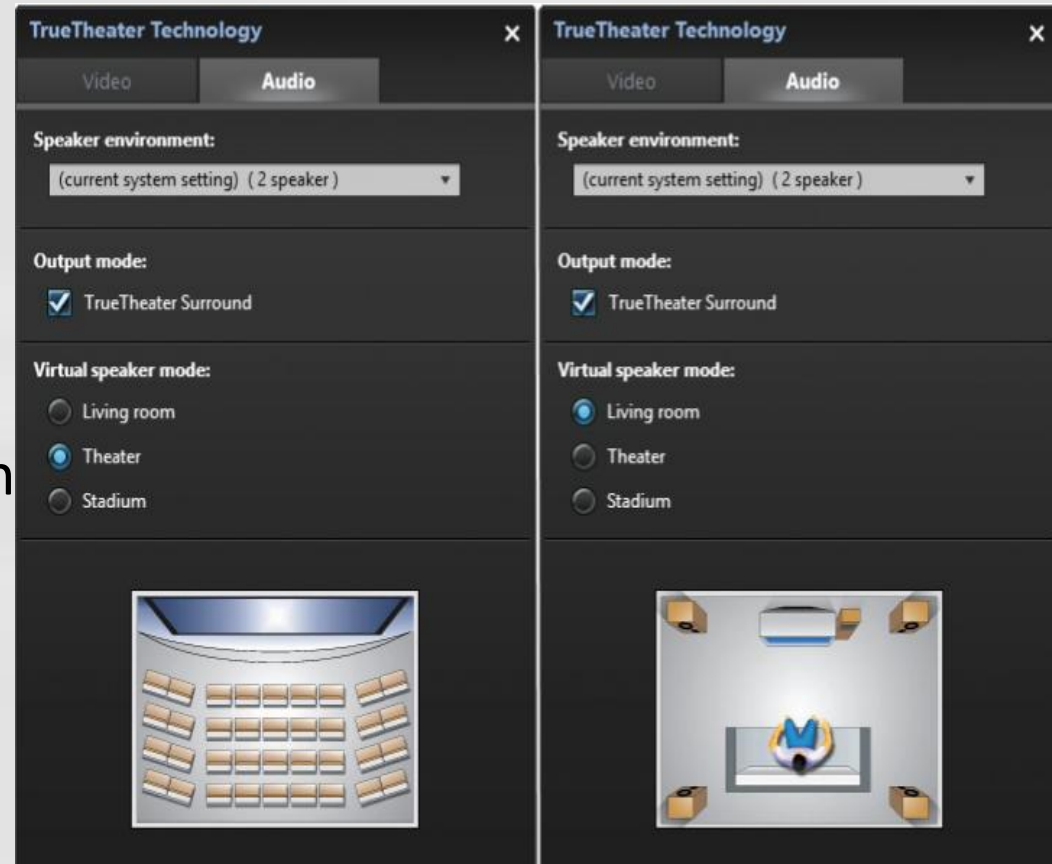
- Human-Interface Objects
 - Affordance – the appearance of the object suggests its function. It makes clear how objects can or should be used



- Affordance is the quality or property of an object that defines its possible uses or makes clear how it can or should be used.

Principles of User-Interface Design

- Human-Interface Objects
 - Visible with Feedback
 - Both visible on the display and provides a response to a user action (feedback)
 - Good examples – radio buttons and check boxes



Principles of User-Interface Design

- Consistency
 - Across platforms
 - Within a suite of applications
 - Within a particular application
- Continuity
 - Consistency across releases over time

Windows Continuity

• Windows 7

• Windows 8

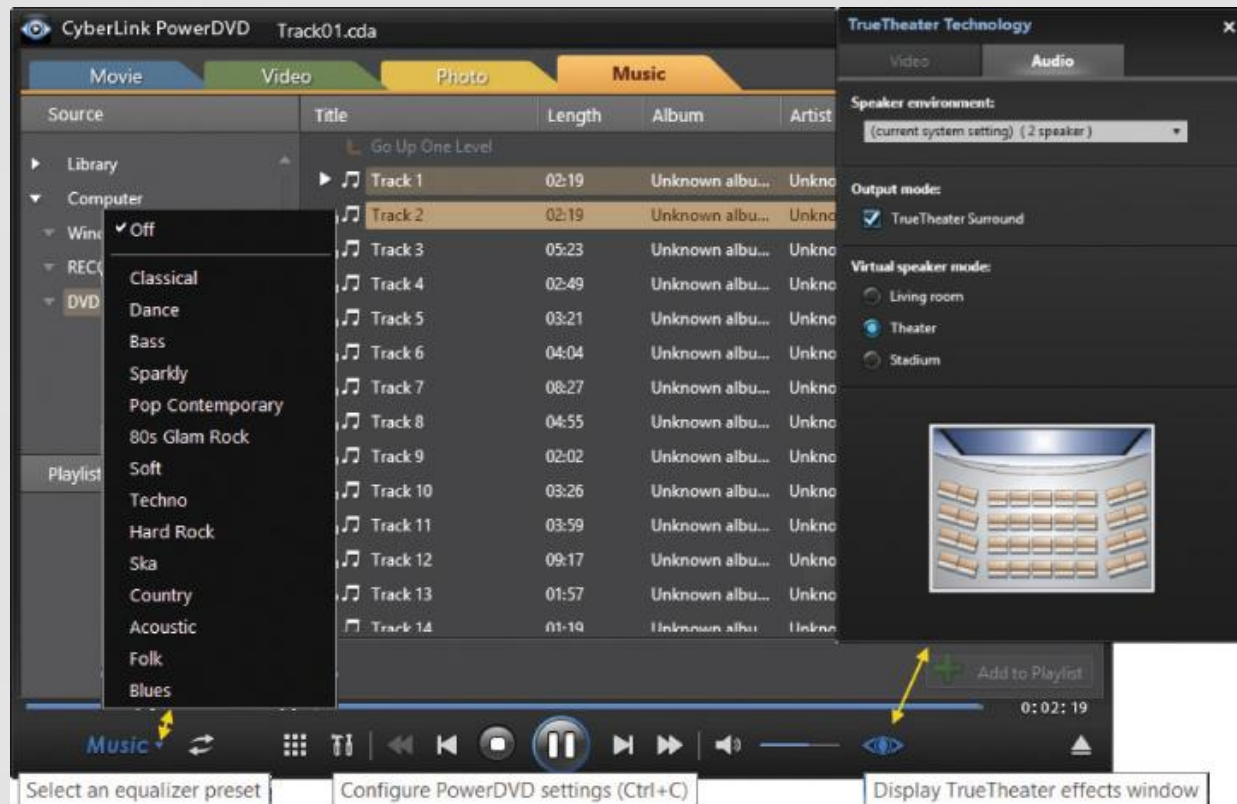
• Windows 10



Principles of User-Interface Design

Discoverability

- To help users discover “hidden” features or objects
- Active discovery – mouse hovers, pop-ups, tool tips



Principles of User-Interface Design

- Discoverability (cont)
 - Visual Diagrams to guide users



Principles of User-Interface Design

- Dialog Closure
 - Closure on Dialogues – End of a series of actions
 - Protect user's work – at end and for partially complete work
 - Provide undo to reverse actions

Principles of User-Interface Design

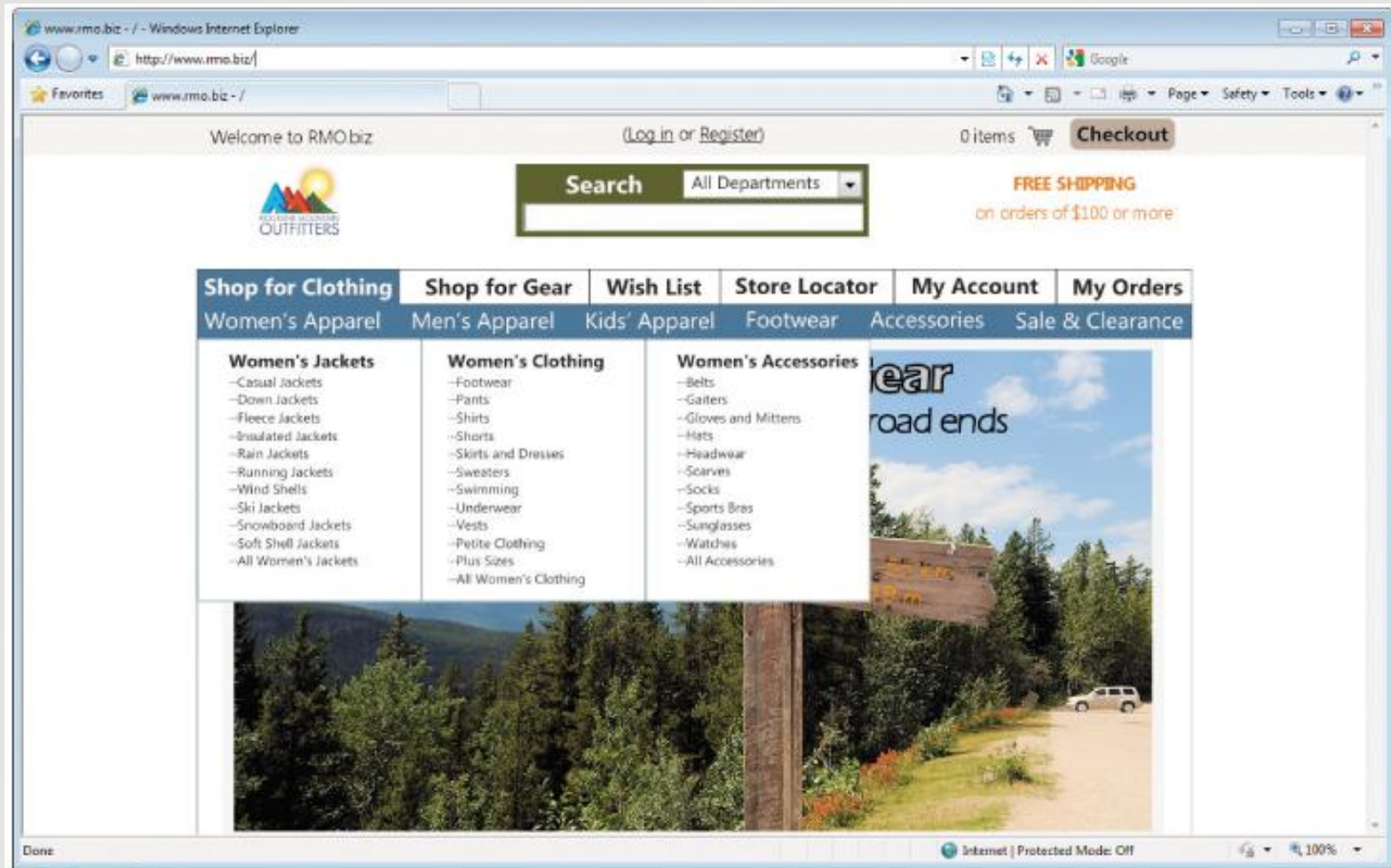
- Readability & Navigation
 - Readable text for all users (type, size, color)



Principles of User-Interface Design

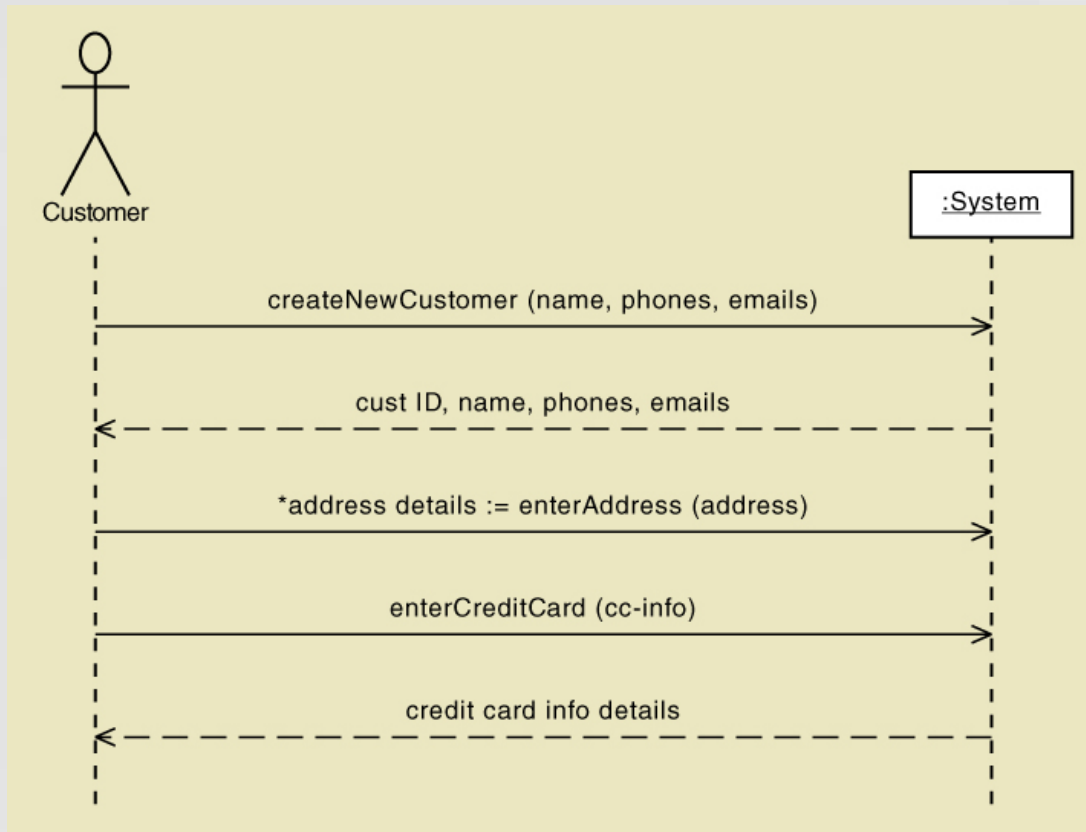
- Readability and Navigation (cont.)
 - Clear navigation
 - Reverse navigation – a way out
- Usability and Efficiency
 - Shortcut keys for experienced users
 - Meaningful error messages
 - Simplicity

RMO Homepage – clean and simple



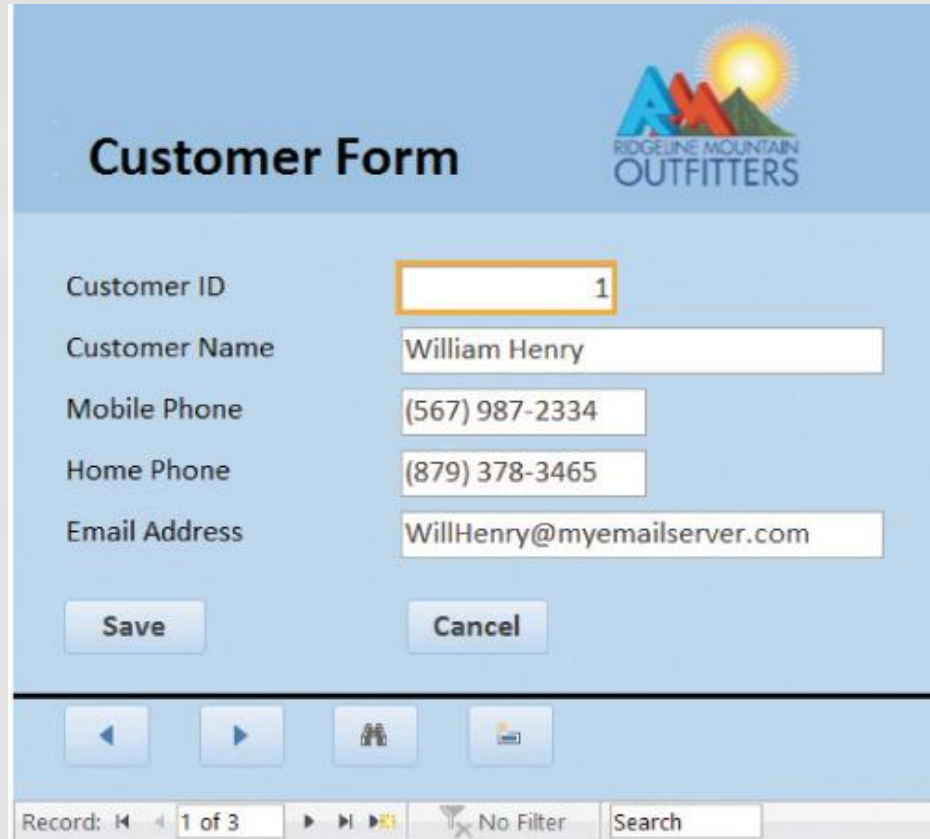
Analysis Models and Input Forms

- SSD defines input messages, which indicates what forms



Sample Customer Form

- First draft of RMO Customer Form from SSD information



The image shows a web-based customer form for Ridgeline Mountain Outfitters. The form has a light blue header with the company logo on the right and the title 'Customer Form' on the left. Below the header, there are five input fields for customer information: Customer ID (containing '1'), Customer Name (containing 'William Henry'), Mobile Phone (containing '(567) 987-2334'), Home Phone (containing '(879) 378-3465'), and Email Address (containing 'WillHenry@myemailserver.com'). At the bottom of the form are two buttons: 'Save' and 'Cancel'. Below the form is a navigation bar with four icons: a left arrow, a right arrow, a group of people, and a document. At the very bottom is a status bar with the text 'Record: 1 of 3', a 'No Filter' button, and a search box.

Customer Form



Customer ID:

Customer Name:

Mobile Phone:

Home Phone:

Email Address:

Navigation:    

Record: 1 of 3

Transitioning from Analysis to UI Design

- Use Cases and the Menu Hierarchy
 - We design use case by use case
 - Menus are a typical way to organize access to use case functionality
 - Different types of users might have different menus
 - Useful to design an overall menu hierarchy and then subsets for different users
 - Once the hierarchy is established, menus can be implemented in a variety of ways

Dialogs and Storyboards

- A storyboard is a graphic organizer that consists of illustrations or images displayed in sequence for the purpose of pre-visualizing a motion picture, animation, or interactive media sequence.
- This is done in systems design to visualize the interactions on a user interface of the application under development (AUD)
- Story boards are powerful tools for UI design

Dialogs and Storyboards

- For each use case, think of the natural flow of a dialog between user and computer
 - Based on the flow of activities in use case description and/or the system sequence diagram
 - Use natural language to emphasize feedback to user
 - Create a storyboard of the dialog, showing the sequence of sketches of the screen each step of the dialog. (storyboarding)
 - Review the storyboard with users

From Dialog to Storyboard (part 1)

Use case *Check out shopping cart*

SYSTEM: What would you like to do?

USER: I'd like to check out.

SYSTEM: Okay. What is your e-mail address or account number?

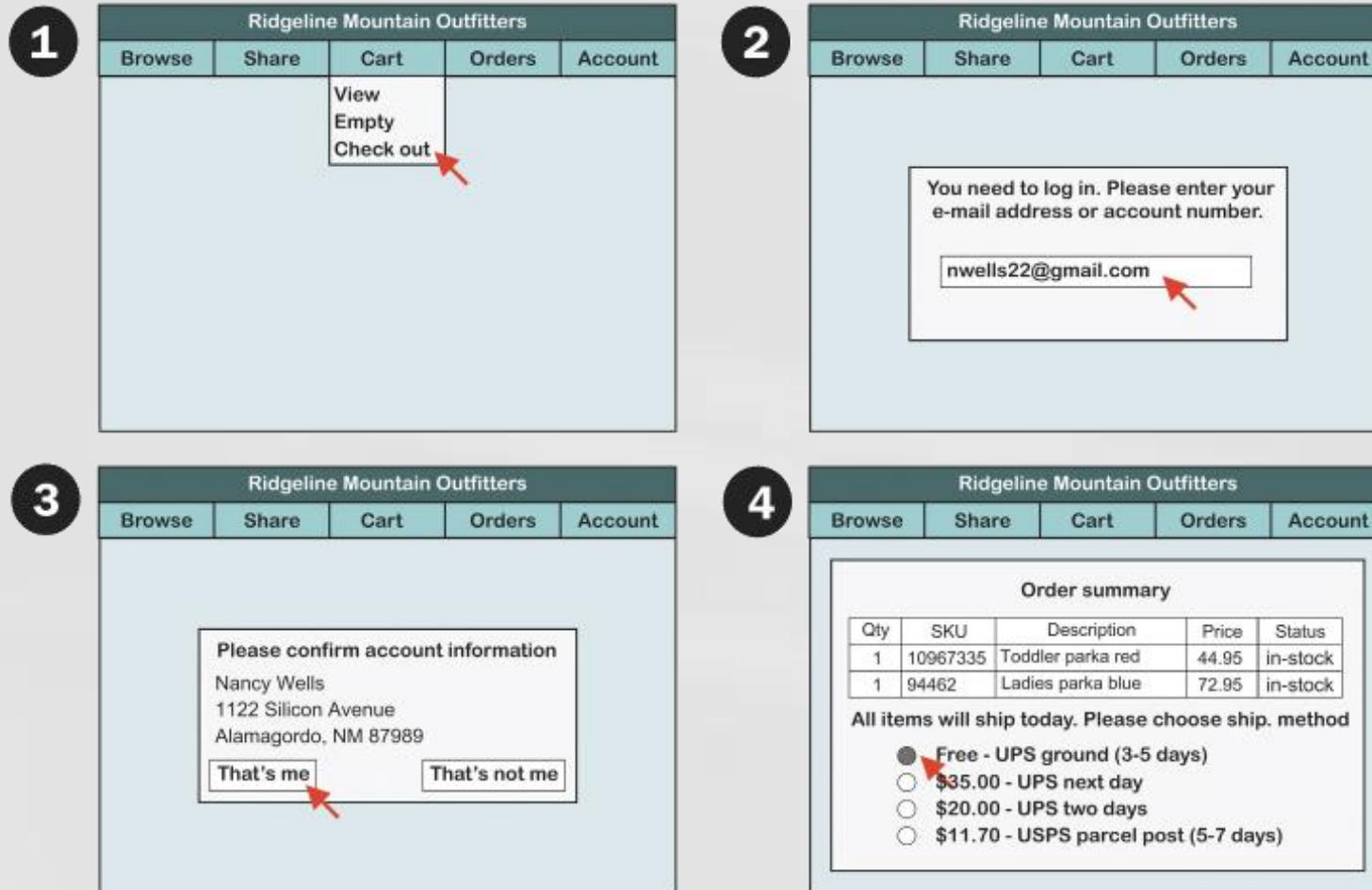
USER: My e-mail address is nwells22@gmail.com.

SYSTEM: Fine. You are Nancy Wells at 1122 Silicon Avenue. Correct?

USER: Yes.

SYSTEM: All items in your cart are in stock and can be shipped today. Because your order subtotal is over \$100, you qualify for free UPS ground shipping (3–5 days). Other shipping options include next day (\$35.00), two day (\$20.00), and USPS parcel post (\$11.70, 5–7 days). How would you like your items to be shipped?

From Dialog to Storyboard (part 1)



From Dialog to Storyboard (part 2)

USER: Free UPS ground shipping.

SYSTEM: We have a shipping address on file for you (1122 Silicon Avenue). Do you want your order shipped to that address?

USER: No.

SYSTEM: To what address would you like the order shipped?

USER: John Wells, 1612 Jefferson Street NE, Albuquerque, NM 87123.

SYSTEM: Okay, the total charge is \$125.56 (\$117.90 item subtotal plus \$7.66 sales tax). Shall I charge that amount to your credit card on file (a Visa with an account number ending in 0899)?

USER: Yes.

SYSTEM: Your payment has been approved, and your order is being prepared for shipment. A confirmation e-mail has been sent to you, and another will be sent with a shipment tracking number when the order is shipped later today. Can I help you with anything else?

USER: No.

From Dialog to Storyboard (part 2)

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Ridgeline Mountain Outfitters

Browse Share Cart Orders Account

Please confirm shipping address

Nancy Wells
1122 Silicon Avenue
Alamagordo, NM 87989

OK Use another address

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Ridgeline Mountain Outfitters

Browse Share Cart Orders Account

Please enter shipping address

Name John Wells
Apt#
Street 1612 Jefferson Street NE
City Albuquerque
State New Mexico
Zip Code 87123

OK Cancel

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Ridgeline Mountain Outfitters

Browse Share Cart Orders Account

Order summary

Qty	SKU	Description	Price	Ext
1	10967335	Toddler parka red	44.95	44.95
1	94462	Ladies parka blue	72.95	72.95
Subtotal				117.90
Shipping				0.00
Sales Tax				7.66
Total				\$125.56

Please confirm payment

Nancy Wells
Visa xxxx-xxxx-xxxx-0899
Exp. 02/17

OK Another method

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Ridgeline Mountain Outfitters

Browse Share Cart Orders Account

Your payment has been approved. Your Visa credit card (xxxx-xxxx-xxxx-0899) has been charged for \$125.56.

Your order number is 6773823.

The order will be shipped today for delivery in 3-5 days.

Thank you shopping with RMO!

User-Interface Design

Some questions to ask

- Is the system a custom application or browser based?
- What kinds of devices will the user-interface need to support?
- What operating systems will the user-interface run on?

User-Interface Design (2 of 3)

- Desktop and Laptop UIs
 - Layout and Formatting
 - Purposeful designs, location and grouping, no sloppiness or errors
 - Poorly designed form –

Payment Options

☒ Pay by Credit Card CardType: Visa Number: ExpDate:

☐ Pay by Debit Card Debit Card: Mastercard Number:

☐ I want to pay by check

☐ Paypal Paypal email:

☐ Send me a bill

Address:

City:

State: Zip:

Interface Design Fundamentals

- Know your user
 - By focusing on your users first, you will be able to create an interface that lets them achieve their goals.
- Pay attention to patterns
 - By using familiar UI patterns, you will help your users feel at home.
- Stay consistent
 - Users need to know that once they learn to do something, they will be able to do it again. Language, layout, are just a few interface elements that need consistency

Interface Design Fundamentals

- Use visual hierarchy
 - Design your interface in a way that allows the user to focus on what is most important. The size, color, and placement of each element helps to create a clear path to understanding your interface
- Provide feedback
 - Your interface should at all times speak to your user, when his/her actions are both right and wrong or misunderstood
 - The interface should provide some feedback during usage thereby letting users know when they have done something unexpected
 - Also when they have done something right

Interface Design Fundamentals

- Be Forgiving

- No matter how clear your design is, people will make mistakes. Your UI should allow for and tolerate user error. Design ways for users to undo actions, and be forgiving with varied inputs (no one likes to start over because he/she put in the wrong date of birth or phone number)

- Empower your user

- Once a user has become experienced with your interface, reward him/her and take off the training wheels. Break down the complex tasks into simple steps. Provide more abstract ways, like keyboard shortcuts etc.

Interface Design Fundamentals

- Speak their language
 - Keep things conversational. Provide clear and concise labels for actions.
- Keep it simple
 - Whenever you are thinking about adding a new feature or element to your interface, ask the question, “Does the user really need this?” or “Why does the user want this?”
- Keep moving forward
 - When creating a UI, you *will* make mistakes. Just keep moving forward to achieve the users expectations.

Video - UX Design

Summary

- User interfaces involve direct user interaction with the system.
- The UI design has a long history as human computer interaction (HCI) and relies on user-centered design, which focuses early on users, evaluates designs to ensure usability, and uses iterative development
- Metaphors are used to think about the nature of the user interface, and they include direct manipulation, desktop, document, and dialog metaphors.

Summary (continued)

- Key user interface concepts include affordance and visibility for controls
- Other key principles include consistency, shortcuts, feedback, dialog closure, error handling, and reversal of actions
- Dialogs and storyboards are used to design the interaction for each use case based on use case flow of activities and system sequence diagrams
- Use case specification is a breakdown of a single use case to provide the description and detailed information about that use case.

Questions?